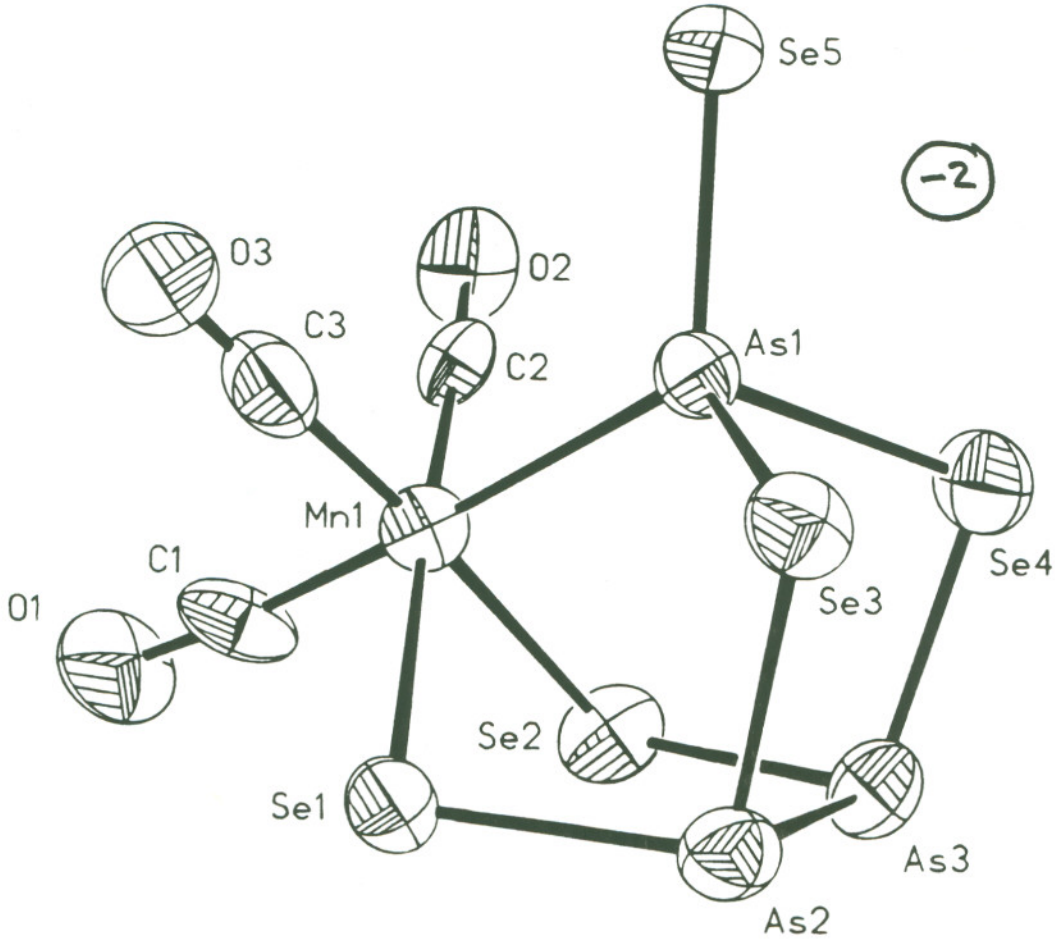
INORGANIC CUMULATIVE EXAM

October 28, 1995

A recent issue of the ACS journal INORG CHEM had an article about the -2 complex ion pictured below.

1. To the first approximation which, if any, of the bonds are likely to have a bond order of greater than one?
2. The As-Se fragment was described as "a -3 ion acting as a 6 electron donor". Explain the logic of that description. (You may want to use an electron dot (Lewis) sketch)
3. What is the "electron count" for the Mn in this complex ion?
4. Give a reasonable preparation scheme: include reagents, possible solvents, and reaction conditions.
5. How many peaks, with what relative intensities, are expected for the ^{13}C NMR?
6. How many peaks, with what relative intensities, are expected for the Se NMR (the only Se isotope with a spin has a spin of 1/2)?
7. Why is the ellipsoid for As1 smaller than that for O1 in this figure?
8. If the As was a better "pi-bonder" than Se in this unit, give the effect on
   (a) the carbonyl stretching frequencies (b) the Mn - C distances (c) the C-O distances
9. What experiment would you do to see if this molecule was fluxional? What would the data look like if it was fluxional? What movements in the complex might make it be "fluxional"?

10-11. Make a suggestion as to what you might try to do as a "next step" in the research in this area. (Be bold in your plan, and explain how it would be really important research).